Label Your Data

with **H2O Label Genie**

H2O.ai Confidential

Label Your Data : Label Genie

H2O.ai

H2O LABEL GENIE

AI assisted data labelling

Intelligent system with zero-shot models

1. Intelligent Annotation
2. Image, Text, Audio
3. Publish Data for Training

H2O.ai Confidential

Label Your Data : Label Genie Potential Use Cases

Task	Benefit
Help to automatically label large volumes of free text responses in surveys so that downstream analysis can be conducted. E.g. Categorise the types of welfare benefits officers want (e.g. F&B vouchers, sport events, Holidays etc.)	The traditional process is tedious and takes time since officers have to read through every single response and manually tag the information. With Label Genie, the tagging is done automatically and users just need to verify and correct the output. This allows users to spend on time on analysis and contributes to a faster turnaround time.

H2O.ai Confidential

Supported Data

Text

Classification

Regression

Entity Recognition

Summarization

Text-generative AI

Audio

Classification

Regression

Images

Classification

Regression

Object Detection

Image instance segmentation

H2O.ai Confidential

Supported Data

Text

- Classification
- Regression
- Entity Recognition
- Summarization
- Text-generative AI

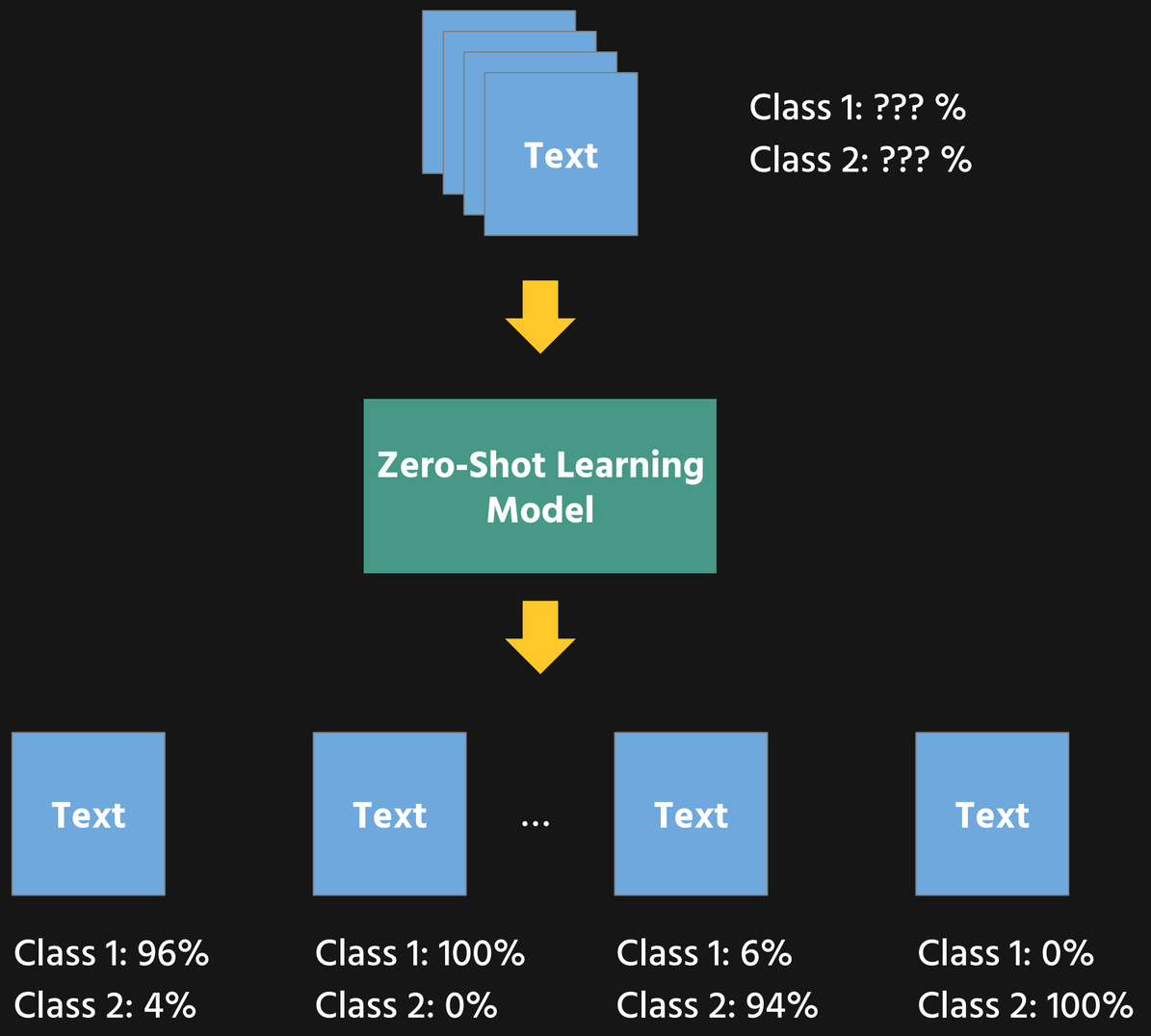
H2O.ai Confidential

Zero-Shot Learning Models Overview

How does "Zero-Shot Learning Models" help?

- Eliminate the need for labelled data upfront (models already trained on vast and varied classes)
- Enable **high-accuracy**, **rapid** labelling without the cost and delay of manual data labelling

Text Classification: Uses the ***bart-large-mnli*** zero-shot learning model by default.



H2O.ai Confidential

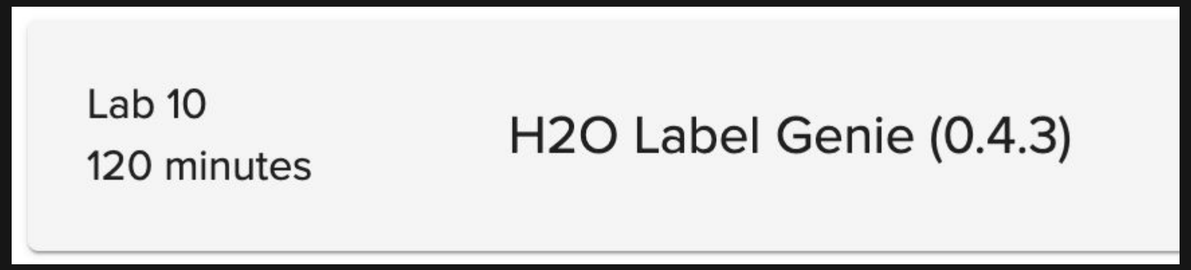
Label Your Data : Label Genie

Platform Access

www.aquarium.h2o.ai

Step 1:
Connect to the Aquarium platform as indicated in the following instructional video from YouTube: Getting Started with H2O.ai Aquarium

Step 2:
Select the Label Genie Lab.



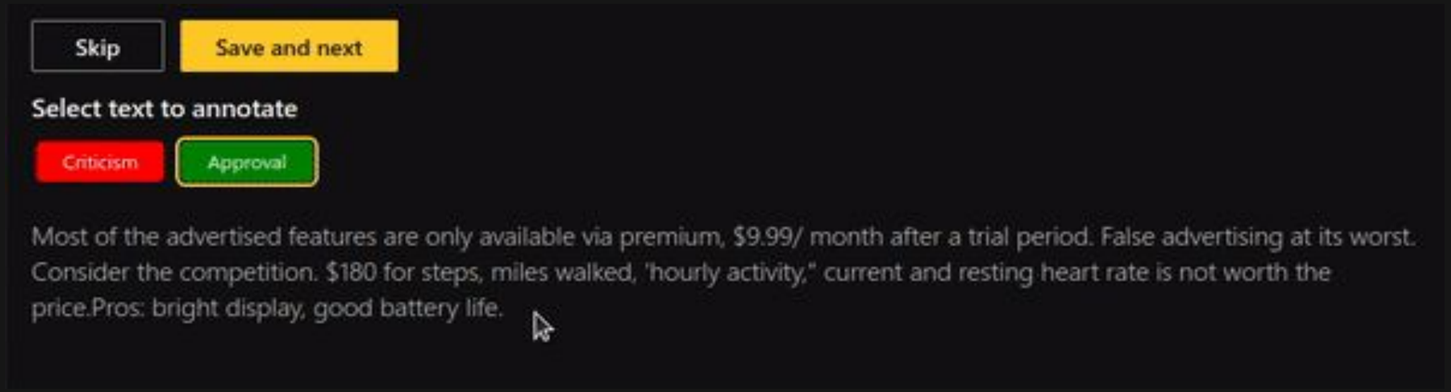
Step 3:
Start the Lab and enjoy!

Label Your Data : Label Genie

1. Platform Access → 2. Labelling → 3. Publishing

Multiple annotation types

- Classification (Image, Audio, Text)
- Regression (Image, Audio, Text)
- Object detection (Image)
- Entity recognition (Text)
- Summarization (Text)



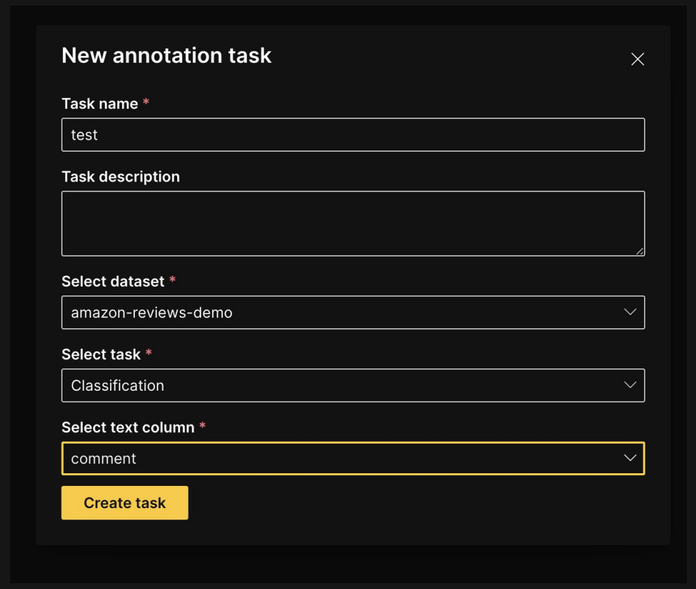
H2O.ai Confidential

Label Your Data : Label Genie

1. Platform Access → 2. Labelling → 3. Publishing

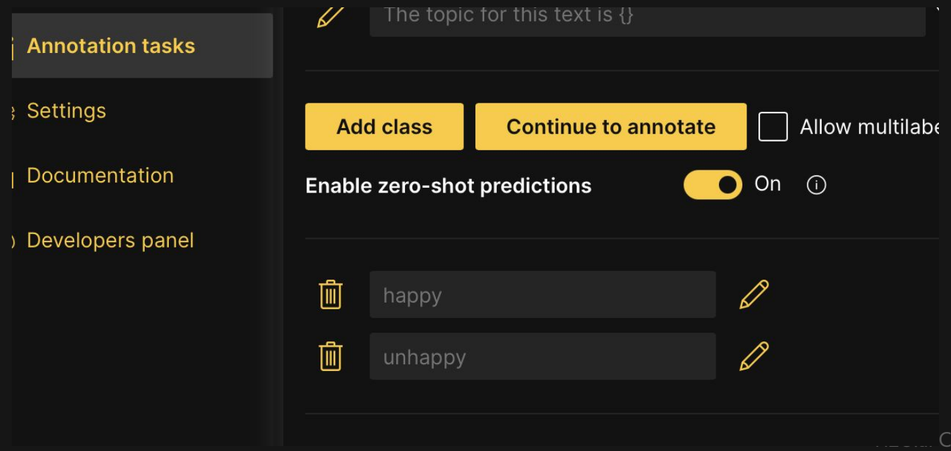
a. Create an annotation task

- "Text Classification" example with Amazon product reviews
- Task list: Classification
- Text column: Comment



b. Specify annotation task rubric

- Class 1: "Happy"
- Class 2: "Unhappy"



c. Use zero-shot labelling + human review

Confidential

Label Your Data : Label Genie

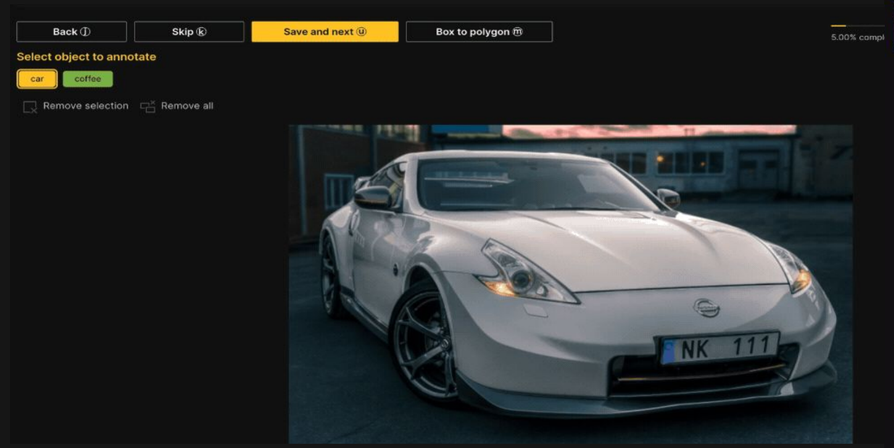
1. Platform Access → 2. Labelling → 3. Publishing

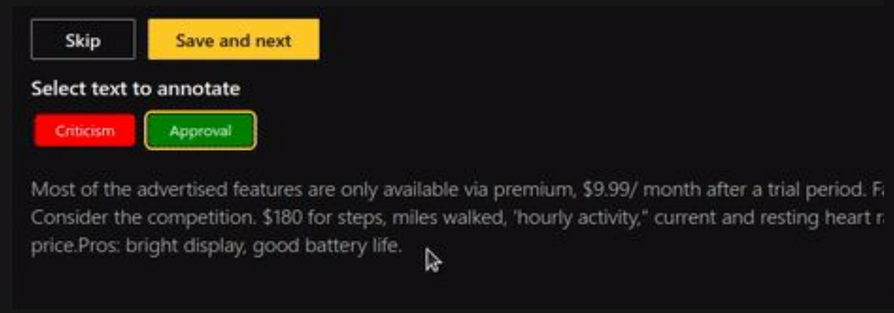
Multiple annotation types

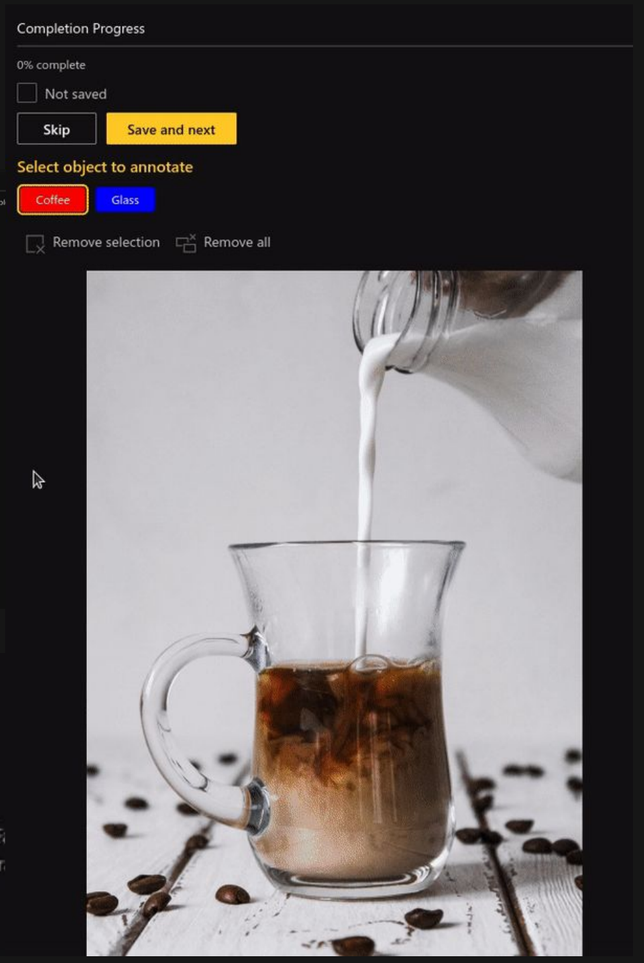
- Classification (Image, Audio, Text)
- Regression (Image, Audio, Text)
- Object detection (Image)
- Entity recognition (Text)
- Summarization (Text)
- Text-generative AI

Multiple annotation types

- Zero Shot Labelling
- Self Learning







H2O.ai Confidential

Label Your Data : Label Genie

1. Platform Access → 2. Labelling → 3. Publishing

- Download the annotated dataset or Export the dataset to H2O Drive

- The downloaded annotated dataset can be used in H2O Hydrogen Torch experiments and H2O LLM Studio.

H2O.ai Confidential

Summary

- **AI-Assisted Data Labelling**
  - Zero-shot learning for intelligent annotation, reducing manual effort
  - Support labelling for text, image, and audio
  - Annotation types include: classification, regression, object detection, entity recognition, and summarisation
- **Workflow**
  - Data -> Define Annotation Rubric -> Annotate -> Export
- **Data Publishing and Export**
  - Export labelled datasets and use them in H2O Hydrogen Torch or H2O LLM Studio for model training

H2O.ai Confidential

H2O.ai

Thank you